

A dichotomous Leggett–Garg witness for braid-representation density in $SU(2)_k$: exact at $d = 2$, dimension-limited at $d \geq 3$

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(Dated: 2026-10-03)

Whether a *temporal* (Leggett–Garg) measurement can certify the computational power of an anyonic braid representation—its *density* in the unitary group, the property underlying universal topological quantum computation—has, to our knowledge, not been studied. We give a complete operational characterization of when the dichotomous Lüders–Leggett–Garg witness $K_3 = 2C(B) - C(B^2)$ (one of several three-time Leggett–Garg witnesses) resolves braid-representation density across the $SU(2)_k$ family. At $d = 2$ (the spin- $\frac{1}{2}$ fusion space) the witness resolves density *exactly*: it saturates the dimension-independent Lüders bound $3/2$ on every dense representation and stays strictly below it on the finite ones, which on this minimal-rank, one-qubit space occur precisely at $k \in \{2, 4, 8\}$ —a sharp threshold distinct from the all-rank finite set $k \in \{1, 2, 4\}$ of Freedman–Larsen–Wang and Kuperberg. The $k = 4$ representation is structurally inert under K_3 at $d = 2$ ($K_3 = 1$ for *every* measurement axis) despite a *larger* quantum dimension than $k = 3$, its braid image being finite; and $k = 8$ is pinned at $3/\sqrt{5}$ (at $Q = \hat{z}$; 1.427 axis-optimized). Both follow from the underlying $SO(3)$ geometry. At $d \geq 3$ the *same* witness ceases to resolve density—it saturates $3/2$ even on finite representations—and we trace this to the extra dimension itself, not to any particular braid angle. This is shown directly, in the state-optimized reading, on three finite $d = 3$ families; in the neutral reading it is established for odd $d \geq 3$, and the neutral case at even $d \geq 4$ is left open. We frame the resulting no-go question—whether *every* such witness is density-blind at $d \geq 3$ —as an open problem.

I. INTRODUCTION

Projective density on the selected fusion space is the representation-theoretic criterion for braid universality on that encoded space [1, 2]. Density is therefore the structural property one would most like to *certify operationally*. Existing operational benchmarks for anyonic computational power are *spatial*: Howard and Vala use a CHSH–Bell violation as a universality benchmark for Ising anyons [5]. The complementary *temporal* route—a Leggett–Garg (LG) test [6, 7], which probes the non-classicality of a *single* system measured sequentially—has, to our knowledge, never been connected to braid-representation density. A companion Letter reports the isolated Fibonacci ($k = 3$, $d = 2$) case of this LG-density connection [16]; the present work is supplemented by that Letter and generalizes the connection to density across the $SU(2)_k$ family.

Here we characterize that connection across the $SU(2)_k$ anyon family, resolved by level k and anyon spin j . We find a clean two-sided result. At $d = 2$ the witness is a faithful density probe with sharp, fully characterized structure; at $d \geq 3$ it stops resolving density, for a reason we pin down. The negative half is not an aside: it converts “we found a witness that works at $d = 2$ ” into “we know exactly where this witness works, where it stops, and why,” and pre-empts the obvious generalization question instead of leaving it open.

We are careful about what is and is not claimed (Sec. VI): the $3/2$ Lüders bound is established prior art [8, 9] which we *cite*, not claim; the operational characterization of where *this* witness tracks density is, to our knowledge, new.

II. SETUP

The witness. For a three-time LG protocol with a dichotomic observable Q ($Q^2 = I$, eigenvalues ± 1) measured under a single braid unitary B , with the *Lüders* (degeneracy-preserving) update rule, the stationary three-time correlator combination studied here is

$$K_3 = 2C(B) - C(B^2), \quad C(U) = \frac{1}{d} \text{Re Tr}[QUQU^\dagger], \quad (1)$$

where d is the dimension of the braid representation (the fusion space) and the prefactor $1/d$ is the maximally mixed state $\rho = I/d$. The symmetrized two-time correlator form $C = \frac{1}{2}\langle\{Q_i, Q_j\}\rangle$ for sequential Lüders measurements of a dichotomic observable is that of the temporal CHSH scenario [10]; our $C(U)$ is its $\rho = I/d$ specialization. For a general state ρ we use the full sequential-Lüders statistics, which yields a witness operator M with $K_3(\rho) = \text{Tr}[\rho M]$; the state-optimized value is $\rho\text{-opt} = \lambda_{\max}(\frac{1}{2}(M + M^\dagger))$ and the maximally-mixed value is $\text{Tr}[M]/d$. K_3 is one of several three-time LG witnesses, which differ in the signs of the three correlators [7]; all LG results in this work refer to K_3 .

The bound. Under Lüders, $K_3 \leq 3/2$ *independently of dimension* [8, 9]; the bound can be exceeded (up to the algebraic value 3) *only* under the degeneracy-breaking

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von-Neumann rule, within the projective measurement framework considered in Ref. [9], which we do *not* use (a different LG inequality, with its own bound, is treated in Ref. [11], Sec. VI). We write “fires” := *saturates* $3/2$. This precise meaning matters: $k = 8$ at $d = 2$ gives $K_3 > 1$ yet does *not* saturate (Sec. III).

The family. $SU(2)_k$ representations are indexed by level k and spin j ; the fusion space of three anyons (chosen total charge) has dimension d , with $j = \frac{1}{2} \rightarrow d = 2$ ($k \geq 2$), $j = 1 \rightarrow d = 3$ ($k \geq 4$), $j = \frac{3}{2} \rightarrow d = 4$, $j = 2 \rightarrow d = 5$. “Resolving density” means: does the witness value distinguish *dense* representations from *finite* ones?

The engine. We use a projective F/R -symbol representation of B_3 (q-deformed $6j$ /Racah), which is phase-blind and therefore sees exactly the *projective* braid image (determinant phase removed)—the density notion relevant to a correlator-based witness. This phase-blindness is appropriate, not a limitation: global phases are physically irrelevant, and a correlator-based witness sees only the projective image in any case. A gate-0 check reproduces the published $d = 2$ values exactly, and the Yang–Baxter relation holds to residual $\lesssim 10^{-15}$ across $d = 2, \dots, 5$ (Appendix A).

The reported witness value. For a given representation, the witness value we report is the supremum of $K_3(B)$ over the braid-group image $\rho(B_3)$, maximized over the measurement axis; “fires” means this supremum reaches $3/2$. It is this supremum over the image—not the value of any single generator—that constitutes the density diagnostic: a *dense* image approaches $3/2$ as the supremum of a never-exactly-attained set, whereas a *finite* image attains only a discrete set of values and can fall strictly short.

III. DENSITY RESOLUTION AT $d = 2$

At $d = 2$ (spin- $\frac{1}{2}$) the witness resolves density, and the resolution is *biconditional in both directions* (Table I, Fig. 1).

TABLE I. K_3 at $d = 2$ (spin- $\frac{1}{2}$) across $SU(2)_k$. Dense levels saturate the Lüders bound $3/2$; the finite levels $k \in \{2, 4, 8\}$ (this minimal-rank, one-qubit space; see Sec. III) stay strictly below. Values are the axis/ Q -optimized K_3 ; the $Q = \hat{z}$ column is shown where it differs.

k	dense?	$ G $	$K_3(Q=\hat{z})$	K_3 (opt.)	sat. $3/2$?
2	no	48	1.000	1.000	no
3	yes	—	—	1.4999	yes
4	no	24	1.000	1.000	no
5	yes	—	—	1.4997	yes
6	yes	—	—	1.4999	yes
7	yes	—	—	1.5000	yes
8	no	120	$3/\sqrt{5} \approx 1.342$	1.427	no
9	yes	—	—	1.4999	yes
10	yes	—	—	1.5000	yes

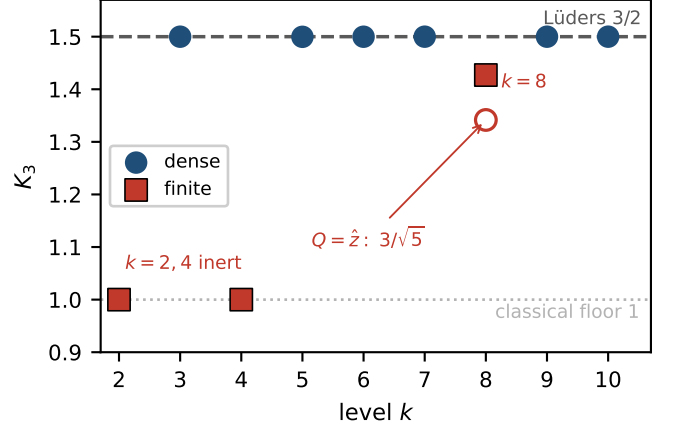


FIG. 1. K_3 at $d = 2$ (spin- $\frac{1}{2}$) across $SU(2)_k$, axis-optimized reading. Legend: “dense”/“finite” refer to the braid image; density on this space is the criterion for braid universality on it. Dense levels $k \in \{3, 5, 6, 7, 9, 10\}$ saturate the Lüders bound $3/2$; the finite levels $k \in \{2, 4, 8\}$ (minimal-rank, one-qubit space) stay strictly below— $k = 2, 4$ at the classical floor 1, and the icosahedral $k = 8$ at 1.427 (its $Q = \hat{z}$ reading $3/\sqrt{5} \approx 1.342$ is the open marker). The closest finite representation, $k = 8$, sits only ≈ 0.07 below the bound.

Forward (dense $\rightarrow$ saturates): the dense levels $k \in \{3, 5, 6, 7, 9, 10\}$ all reach $3/2$ to the precision shown—the values approach $3/2$ as the supremum of a dense (never exactly attained) image, with finite-word/sampling estimates landing at 1.4997 – 1.4999 , while $k = 7, 10$ sit at $3/2$ to machine precision ($|3/2 - K_3| \lesssim 10^{-14}$). *Backward* (saturates $\rightarrow$ dense, i.e. finite reps stay strictly below): the finite levels are exactly $k \in \{2, 4, 8\}$, with $SU(2)$ braid images the binary octahedral group ($k = 2$, order 48), the binary tetrahedral group ($k = 4$, order 24), and the binary icosahedral group ($k = 8$, order 120)—equivalently $r = k + 2 \in \{4, 6, 10\}$, the finite set at this minimal rank identified by Freedman–Larsen–Wang [2], with the underlying B_3 representation-theoretic machinery due to Tuba–Wenzl [3] (classification of B_3 representations) and Rowell–Tuba [4] (finite-image criterion). We emphasize that $\{2, 4, 8\}$ is the finite-image set specific to the $d = 2$ (one-qubit, B_3) fusion space considered here. This differs from the all-rank finite set $k \in \{1, 2, 4\}$ ($r \in \{3, 4, 6\}$) of Freedman–Larsen–Wang and Kuperberg [1, 2]: $k = 1$ ($r = 3$) gives a one-dimensional representation, excluded from a qubit setting, while $k = 8$ ($r = 10$) is the icosahedral borderline case—finite at minimal rank, hence finite here at $d = 2$, but dense at higher rank, which for $r = 10$ is established at $n \geq 5$ by Freedman–Larsen–Wang [2]. On the $d = 2$ space the finite set is therefore $\{2, 4, 8\}$, not $\{1, 2, 4\}$; both statements are correct on their respective domains. Since the engine is phase-blind it realizes the projective ($PSU(2) \cong SO(3)$) images, of orders 24, 12, 60 for $k = 2, 4, 8$; the orders $|G| = 48, 24, 120$ quoted above are the binary ($SU(2)$) lifts. The identification is fixed geometrically: σ_1 acts as a 90° (4-fold) rotation at

$k = 2$ —forcing the octahedral group, since the tetrahedral group has rotation angles $\{0^\circ, 120^\circ, 180^\circ\}$ only—as a 120° rotation at $k = 4$ (tetrahedral), and as a 144° rotation at $k = 8$ (icosahedral). These three angles are the finite-level instances of the general sector-phase family $\delta^*(k) = \pi k/(k+2)$ ($k = 2 \rightarrow 90^\circ$, $k = 4 \rightarrow 120^\circ$, $k = 8 \rightarrow 144^\circ$); the $\sigma_1 = 144^\circ$ identification above is exactly $\delta^*(8)$. Their witness values stay below $3/2$: $k = 2, 4$ at 1.000, and $k = 8$ at $3/\sqrt{5} \approx 1.342$ ($Q = \hat{z}$) or 1.427 (optimal)—both $< 3/2$. This two-sided hold is the substance of the $d = 2$ result, and (Sec. IV) the backward direction is exactly what no longer holds at $d \geq 3$.

As a corollary worth stating: at $d = 2$ the witness operationally separates the *universal* Fibonacci model ($k = 3$, dense—its K_3 approaching $3/2$ as the supremum of a dense image) from the *nonuniversal* finite levels $k \in \{2, 4, 8\}$ (this minimal-rank space; Fibonacci anyons are $SU(2)_3$; this separation is a $d = 2$ phenomenon, lost at $d \geq 3$).

Sharp finite set and the $k = 4$ surprise. The finite set on this minimal-rank space is precisely $\{2, 4, 8\}$; in particular $k = 4$ is not merely below $3/2$ but *structurally inert* under K_3 : $K_3 = 1$ for *every* measurement axis, established by a closed argument $\max_U \lambda_{\max}(M(U)) = 1$ (an eigenvalue argument, not sampling; 0 of the 24 group elements violate). This is counterintuitive: $k = 4$ sits at the floor *despite a larger quantum dimension than $k = 3$* ($D_4 = \sqrt{3} \approx 1.732 > D_3 = \varphi \approx 1.618$, with $D_{1/2} = 2 \cos[\pi/(k+2)]$)—its braid image being *finite* (order 24). Large quantum dimension does not predict the witness response; density (here, its absence) does.

The $k = 8$ value (closed form). $K_3(k=8, \hat{z}) = 3/\sqrt{5}$ is proven in closed form from the icosahedral structure: with $K_3 = f(\theta, p) = 1 + 2c - 2c^2 - 2pc(1-c)$ ($c = \cos \theta$, $p = n_z^2$), the identity $f(c, 1) = 1$ (the $2c - 2c^2 = 2c(1-c)$ term cancels the $-2pc(1-c)$ term exactly at $p = 1$, for every θ) kills all $\theta \notin (0^\circ, 90^\circ)$, leaving only the A_5 angle 72° ; the minimizing axis overlap $n_z^2 = 1/5$ is the icosahedral 5-fold $\leftrightarrow$ 5-fold angle ($\cos = 1/\sqrt{5}$). Here $\hat{z}$ is taken along a 5-fold axis of the icosahedral image—witnessed by σ_1 , which acts as a 144° rotation about $\hat{z}$. A rotation about the measurement axis has $n_z^2 = 1$, so by the identity $f(c, 1) = 1$ it contributes $K_3 = 1$ regardless of angle; in particular σ_1 itself gives $K_3 = 1$ in this frame, *not* $3/\sqrt{5}$. The value $3/\sqrt{5}$ is instead realized by a distinct group element—a 72° rotation about a *different* 5-fold axis, sitting at the icosahedral 5-fold $\leftrightarrow$ 5-fold separation $\cos^{-1}(1/\sqrt{5})$ from $\hat{z}$ ($n_z^2 = 1/5$). Among the A_5 rotation angles $\{72^\circ, 120^\circ, 144^\circ, 180^\circ\}$, only 72° lies in $(0^\circ, 90^\circ)$, and $f(72^\circ, 1/5) = +3/\sqrt{5}$ is the largest value attained at fixed $Q = \hat{z}$ over the group; the 144° element about the same tilted axis gives the negative branch $f(144^\circ, 1/5) = -3/\sqrt{5}$.

Correction to v1.0: the first release associated the $K_3 = 3/\sqrt{5}$ maximum with the 144° generator rotation about the measurement axis. As derived above, rotations about the measurement axis yield $K_3 = 1$ for every angle; the maximum instead arises from the 72° rotation about

a tilted five-fold axis ($p = 1/5$). No numerical value is affected.

Mechanism (structural, not topological). $C(U)$ reduces to the zz -component of the $SO(3)$ rotation $R(U)$, so K_3 is an $SO(3)$ -geometric quantity; $k = 8 = 3/\sqrt{5}$ follows from icosahedral geometry, and the $k = 4$ floor from the absence of any tetrahedral-group rotation in $(0^\circ, 90^\circ)$. This is representation theory / finite-subgroup geometry—not topology.

IV. LOSS OF RESOLUTION AT $d \geq 3$

At $d \geq 3$ the same witness no longer separates dense from finite under the targeted reading, and under the neutral reading for odd $d \geq 3$; for even $d \geq 4$ the neutral case is left open (Fig. 2).

(a) *Neutral ($\rho = I/d$) under-resolves.* The maximally-mixed value has the closed form $(3d-1)/(2d)$ for odd d and $3/2$ for even d . For odd $d \geq 3$ this cap is *below* $3/2$ ($d = 3 \rightarrow 4/3$, $d = 5 \rightarrow 7/5$), and *both dense and finite* representations reach it—e.g. at $d = 3$ the dense $k = 5, \dots, 10$ and the finite $(4, 1)$ all sit at $4/3$. No separation. This neutral no-resolution result is established for *odd* $d \geq 3$; for even $d \geq 4$ the neutral cap is again $3/2$, as at $d = 2$, and whether the neutral reading resolves density there is left open. The targeted reading below over-resolves at every $d \geq 3$ we tested, even and odd alike.

(b) *Targeted (ρ -opt) over-resolves.* The state-optimized value reaches $3/2$ for the braid dynamics of the $SU(2)_k$ family at every $d \geq 3$ we tested (Fig. 2). Crucially it does so *even for finite (nondense) representations*. We verified this on three independent finite $d = 3$ families (Table II). Since dense $d = 3$ reps also reach 1.500 under ρ -opt, firing no longer discriminates density. (All three live in the smallest nontrivial total-charge sector; the apparent $d = 3$ -vs- $d = 5$ multiplicity at $(8, 2)$ is resolved in Appendix B.)

All three directly-tested finite representations live at $d = 3$; no finite $d = 5$ representation was checked directly. No finite $d = 5$ anyon representation exists to test in the level range swept ($k \leq 16$): the $(8, 2)$ representation at $d = 5$ is itself dense. For $d \geq 5$ the result therefore rests on the closed-form neutral cap $(3d-1)/(2d)$ (computed through $d = 7$) together with the dimension mechanism below—*shown* on three finite $d = 3$ reps and structurally expected, rather than directly verified, at higher d .

TABLE II. The targeted witness fires ($K_3 = 3/2$) on *finite* $d = 3$ representations, verified on three independent families.

finite rep (k, j)	$ G $	targeted K_3	fires?
$(4, 1)$	162	1.500	yes
$(8, 2)$	450	1.500	yes
$(12, 3)$	882	1.500	yes

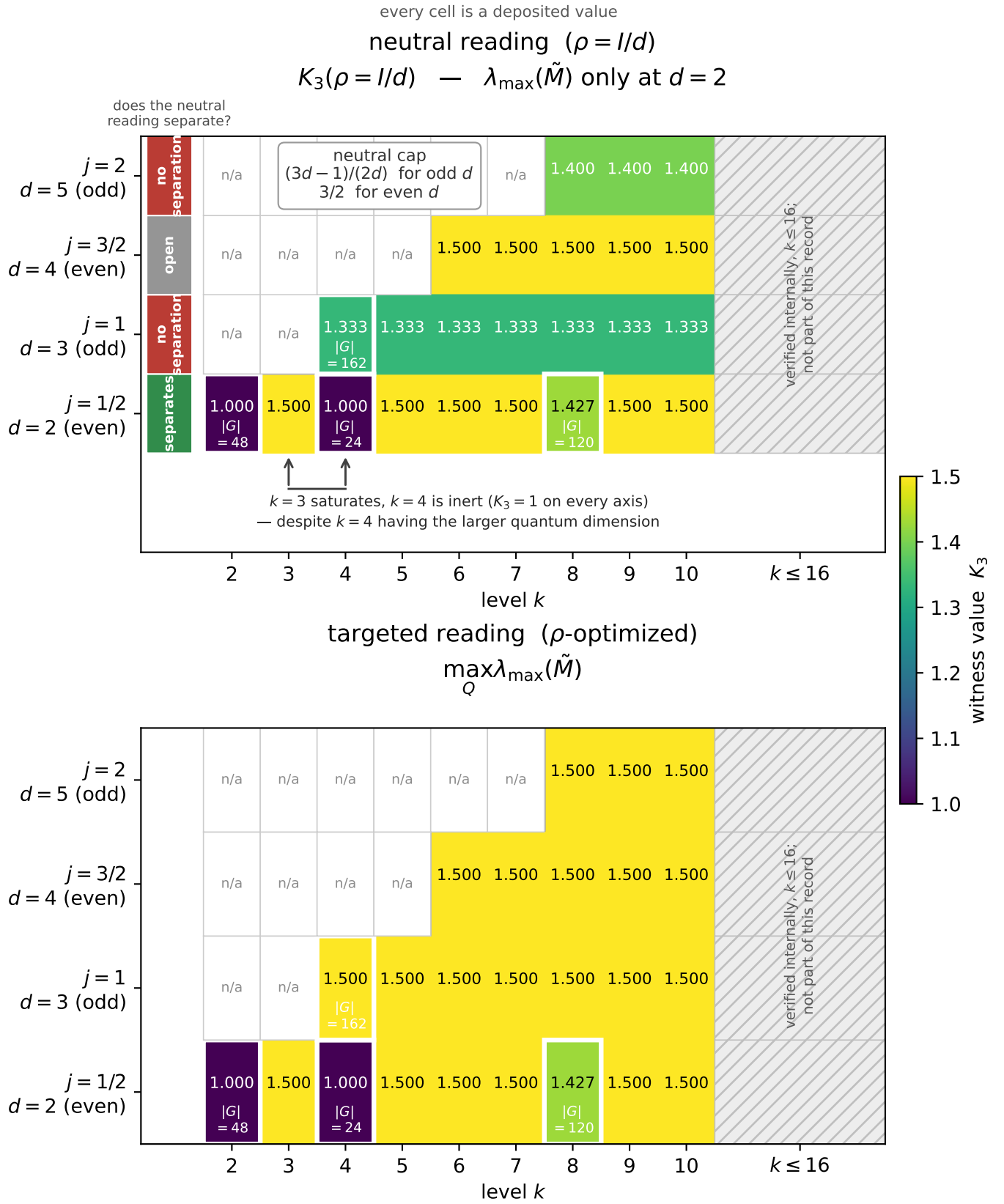


FIG. 2. Resolution map across the $SU(2)_k$ anyon family, by level k and anyon spin j . Left strip: does the neutral reading separate dense from finite? Every cell is a deposited value—the targeted panel and the $j = \frac{1}{2}$ ($d = 2$) row of the neutral panel from `p6_gate1_sweep_results.json`, the neutral values at $d \geq 3$ from the closed form in `p6_rho_closed.json`; the hatched column marks levels verified internally but not part of this record.

TABLE III. Dimension nail: the same rotation $R_y(\theta)$ at $d = 2$ versus $R_y(\theta) \oplus 1$ at $d = 3$. The targeted witness is angle-dependent at $d = 2$ but fires for every angle at $d = 3$.

θ	$d = 2$	$d = 3 \oplus \text{triv}$
60°	1.500 (fires)	1.500 (fires)
72°	1.427	1.500 (fires)
90°	1.000	1.500 (fires)
100°	1.000	1.500 (fires)

State-dependence appears only at $d \geq 3$. At $d = 2$ the neutral and targeted readings *coincide* (state-independent: $k = 4 \rightarrow 1.0/1.0$, $k = 8 \rightarrow 1.427/1.427$); they *split only from $d \geq 3$ on*. The witness change between the $d = 2$ and $d \geq 3$ analyses is therefore not a hidden assumption swap—at $d = 2$ there is nothing to swap.

Mechanism: the extra dimension, not a braid angle. Take the *same* one-parameter rotation $R_y(\theta)$ and compare it as a $d = 2$ unitary versus the same rotation embedded in $d = 3$ as $R_y(\theta) \oplus 1$. The targeted witness is angle-dependent at $d = 2$ but angle-independent at $d = 3$ (Table III)—two disjoint code paths (an algebraic G_{eff} form and a direct sequential-Lüders solver) agree exactly. The extra dimension supplies the spectral freedom to reach 3/2 regardless of the braid angle. This refutes the tempting reading that the witness merely re-uses a $d = 2$ -optimal $\sim 60^\circ$ rotation inside a sub-block: at $d = 3$, $72^\circ/90^\circ/100^\circ$ fire equally. It is the dimension, not the angle.

V. DISCUSSION

The dichotomous Lüders-LG witness K_3 is a faithful density probe *only at $d = 2$* , where its value tracks the projective braid image exactly, with a sharp threshold, a structurally inert $k = 4$, and a closed-form $k = 8$. From $d \geq 3$ it is *scope-limited, not broken*: the targeted reading has too much headroom (the extra dimension lets even finite reps saturate), and for odd $d \geq 3$ the neutral reading has too little (its cap drops below 3/2 and everything reaches it); for even $d \geq 4$ the neutral cap is again 3/2 and the question is left open. The statement is about the *witness*, not about the representations—the density question itself remains perfectly well-posed at $d \geq 3$ and is the natural next target. K_3 is computed here for projective Lüders measurement with the state readings labeled throughout ($\rho = I/d$ neutral or ρ -optimized targeted); its behavior under weak or nonprojective measurement protocols is not addressed and is left open.

VI. RELATION TO PRIOR WORK

The dimension-independence of the Lüders bound 3/2 is established prior art—Budroni, Moroder, Kleinmann,

and Gühne [8] prove 3/2 optimal for all dimensions, and Budroni and Emary [9] establish the Lüders-versus-von-Neumann distinction (dichotomic / $M = 2 \rightarrow 3/2$; full degeneracy-breaking / $M = N \rightarrow \text{exceedable}$). We *cite* this bound; we do not claim it. Ref. [11] treats the *four-term* LG inequality (classical bound 2) and reaches $2\sqrt{2}$ for arbitrary spin via a Gisin-Peres dichotomization; this is *not* a statement about the three-term Lüders bound 3/2 and must not be read as “3/2 exceedable under our convention.” Two experiments corroborate that the Lüders 3/2 is the dividing line: a trapped-ion measurement of $K_3 = 1.739 \pm 0.014$ [12] and an NV-center measurement of $K_3 = 1.625 \pm 0.022$ [13]—the former explicitly under the von-Neumann update rule, the latter via ideal negative-result measurement—both exceeding the Lüders bound. The density classification (finite exactly at $k \in \{2, 4, 8\}$ on this minimal-rank space) follows from Freedman-Larsen-Wang [2]; the $k = 8$ (icosahedral, $r = 10$) borderline case, finite here but dense at higher rank, falls under the same theorem at $n \geq 5$. For the two-eigenvalue / higher- j projective density generally we also rely on Freedman-Larsen-Wang [2]. We read off this classification; we do not establish it.

Two recent works are the closest neighbors and are distinct from ours. Vieira, Milz, Vitagliano, and Budroni [14] witness the *environment* dimension from below via a semidefinite-programming hierarchy—a different witnessed quantity, a different functional, and a different update model; our result is not a special case of theirs. Chou [17] constructs *spatial* KCBS contextuality on Fibonacci / $SU(2)_k$ braid representations—no temporal LG test, no density biconditional.

To our knowledge the bridge “temporal LG witness $\leftrightarrow$ braid density” is unoccupied, and the limitation statement is unpublished. We therefore state, narrowly: this is the first operational characterization that *this* dichotomous Lüders-LG witness resolves projective braid density at $d = 2$ and ceases to do so at $d \geq 3$. We do *not* claim that no temporal witness can resolve $d \geq 3$ density—that generalization is neither proven nor prior-art-cleared (Sec. VII).

VII. OUTLOOK

The natural next question is a genuine *no-go*: is *every* ρ -optimized dichotomous Lüders-LG witness density-blind at $d \geq 3$? If so, it would close an entire class of temporal density probes—a stronger statement than the present result, requiring its own prior-art pass and proof. We pose it as open.

Operational and hardware outlook. Operationally, the witness is a *characterization* tool: numerically it certifies whether a given braid representation is dense—the criterion for universality on the encoded space—and at $d = 2$ it does so faithfully, including for Fibonacci ($k = 3$). A realization on a physical anyon platform would require sequential Lüders measurements (distinct from the spatial

CHSH protocols used in existing anyon-QC benchmarks), synthesis of a degeneracy-preserving mid-circuit measurement, and sufficient resolution against noise given the narrow finite-versus-dense gaps (the closest finite representation, $k = 8$, sits only ≈ 0.07 below the bound in the axis-optimized reading; Fig. 1); we leave this to future work. The values reported here are those of the idealized sequential Lders protocol; an experimental test would additionally have to address the clumsiness loophole—that a violation could be caused by the measurements disturbing the system [7, 18]. The witness certifies the *density of the model*—the criterion for universality on the encoded space—not the correctness of any particular computation.

VIII. CONCLUSION

We have completely characterized when this dichotomous Lders-LG witness resolves projective braid density across the $SU(2)_k$ family: exactly at $d = 2$ (sharp, minimal-rank threshold $k \in \{2, 4, 8\}$; $k = 4$ inert under K_3 for all axes despite the larger quantum dimension; $k = 8$ pinned at $3/\sqrt{5}$ at $Q = \hat{z}$, 1.427 axis-optimized), and not at $d \geq 3$, where the extra dimension—not any braid angle—supplies the spectral freedom to saturate $3/2$ even on finite representations. The $d \geq 3$ side is shown directly, in the state-optimized reading, on three finite $d = 3$ families; in the neutral reading it is established for odd $d \geq 3$, with even $d \geq 4$ left open.

DATA AND CODE AVAILABILITY

Paper, figures, and data are released under the Creative Commons Attribution 4.0 International (CC BY 4.0) license; the deposited source code is released under the Apache License 2.0 (LICENSE and LICENSE-CODE in the record; <https://creativecommons.org/licenses/by/4.0/>). All files backing the tables and figure of this preprint are deposited with this record [15]. The computations use `numpy` and `scipy` (plus `matplotlib` for the figure); random seeds are fixed in the released scripts, so reproduction is deterministic. The correspondence table is deposited as `p6_gate1_sweep_results.json`, and the braid representations are built by `p6_engine.py`; the deposit README lists every script and result file. Gate-0 and Yang-Baxter sanity checks at the head of the engine must pass at machine precision; if they do not, the symbol conventions have been altered. The internal $k \leq 16$ level sweep supporting the $d \geq 5$ scope statement of Appendix C is not part of this record. The reported optima are given as explicit braid words in the deposited results file and can be checked with the included verifier `p6_gate1_verify.py`; the search infrastructure that produced them is not part of this record.

USE OF AI TOOLS

In the research, processing, and writing of this paper and its results I worked together with generative AI tools, in practice a system of multiple coordinated AI instances that I set up and orchestrate (large language models, mainly Claude, by Anthropic, inside Claude Code). At their current context sizes I found it far more effective to work with several specialized instances, each with its own role and its own harness of rules and parameters that I designed and refined through feedback, than to load a single instance with all of the material; for my workflow that would have been inefficient, though this depends on the individual implementation. I lead this collaboration: I choose the research directions, set the goals, and make the final decisions in open exchange with the AI, learning actively as the work proceeds. The AI carries out the drafting, including the mathematical and technical parts, the numerical computation, and the literature search, under my direction. The AI works autonomously only task by task, within the structure I develop through feedback: it completes a task, and at open questions that need me it stops until the point is settled before the next step. Along the way I witness and take many of the decisions that shape the path, and it is common for me to spot things that need improvement. The work spans many separate runs, and a single simulation or build task alone can take up to an hour, so it could not happen all together in one autonomous run; and had I let the AI do all of it together alone, even if it is possible, it would no longer be my work but the AI's.

I run multiple verifications at the different stages of the work and one before release, including cross-checks with an unrelated AI model from a different company, and all references are checked against the original sources. In the end what matters are human eyes, a principle that is itself written into the parameters of my system: I reach out to experts after publishing for review and feedback, so I learn what is solid and what must be corrected or falsified. My scripts for reproduction and review are released with this record. These tools are not authors; I am the author, and I take full responsibility for all scientific content and decisions leading to these results and their publication.

Appendix A: Engine

The braid group B_3 is represented on the d -dimensional fusion space by the projective F/R -symbol (q-deformed $6j$ /Racah) representation. It is phase-blind—the determinant phase is removed—so it realizes the *projective* braid image, the density notion a correlator-based witness can resolve. Validation: a gate-0 check reproduces the published $d = 2$ values exactly, and the Yang-Baxter relation holds to residual $\lesssim 10^{-15}$ across $d = 2, \dots, 5$.

Appendix B: Fusion-charge sectors, and the apparent $d = 3$ -vs- $d = 5$ multiplicity at $(8, 2)$

For three anyons of spin j (label $a = 2j$) the fusion-space dimension depends on the total-charge sector (label $2J$): $d(J) = \sum_i N_{jj}^i N_{ij}^J$ under the level- k truncation. The three finite reps that fire the targeted witness (Table II) all live in the smallest nontrivial sector $2J = 2$, where $d = 3$ (Table IV). This resolves an apparent multiplicity: a sweep over (k, j) lists $(8, 2)$ at $d = 5$ —its *largest* sector ($2J = 4$), which is *dense*—whereas the finite firing representation is the *same* $(8, 2)$ in the $2J = 2$ sector ($d = 3$, finite, order 450). These are different total-charge sectors of one anyon content, not a contradiction. Only $(8, 2)$ produces this surface multiplicity: $(4, 1)$ has $d = 3$ as its only nontrivial sector, and $(12, 3)$ lies outside the sweep’s range.

Appendix C: Closed forms and the $d \geq 5$ scope

$k = 8$, $Q = \hat{z}$. $K_3 = 3/\sqrt{5}$. With $K_3 = f(\theta, p) = 1 + 2c - 2c^2 - 2pc(1 - c)$ ($c = \cos \theta$, $p = n_z^2$), the identity $f(c, 1) = 1$ removes every angle outside $(0^\circ, 90^\circ)$; the only A_5 rotation angle there is 72° , and the icosahedral 5-fold $\leftrightarrow$ 5-fold overlap fixes $n_z^2 = 1/5$, giving $3/\sqrt{5}$. Here $\hat{z}$ is a 5-fold axis witnessed by σ_1 (144° about $\hat{z}$), which itself gives $K_3 = 1$ (not $3/\sqrt{5}$) since $n_z^2 = 1$ for a rotation about the measurement axis; $3/\sqrt{5}$ is realized by the distinct 72° element about the tilted 5-fold axis at $n_z^2 = 1/5$, while the 144° element about that same tilted axis gives the negative branch $-3/\sqrt{5}$ (Sec. III).

Caps. The maximally-mixed value is $(3d - 1)/(2d)$ for odd d and $3/2$ for even d ; the state-optimized cap is $3/2$ for every $d \geq 3$.

$k = 4$ inertness under K_3 at $d = 2$. For a rotation by θ about axis $\hat{n}$, $M(U) = (1 + 2c - 2c^2)I - 2c(1 - c)\hat{n}\hat{n}^\top$, with eigenvalues $\{1, 1 + 2c - 2c^2 (\times 2)\}$; the second exceeds 1 only for $\theta \in (0^\circ, 90^\circ)$, and the tetrahedral group A_4 has rotation angles $\{0^\circ, 120^\circ, 180^\circ\}$ only, so $\lambda_{\max}(M(U)) = 1$ for every axis.

Scope ($d \geq 5$). Every finite braid representation tested directly lies at $d = 3$ (three of them); no finite $d = 5$ anyon representation exists to test for $k \leq 16$ (a level sweep finds only dense $d = 5$ images, e.g. $(8, 2)$). For $d \geq 5$ the conclusion therefore rests on the closed-form cap above (computed through $d = 7$) together with the extra-dimension mechanism—structurally expected, not

TABLE IV. Fusion-space dimension by total-charge sector $\{2J : d\}$ for the three finite families, and the finite $d = 3$ representation each contributes at $2J = 2$.

(k, j)	sectors $\{2J : d\}$	finite $d = 3$ at $2J = 2$
$(4, 1)$	$\{0:1, 2:3, 4:1\}$	order 162
$(8, 2)$	$\{0:1, 2:3, 4:5, 6:3, 8:1\}$	order 450
$(12, 3)$	$\{0:1, 2:3, 4:5, 6:7, 8:5, 10:3, 12:1\}$	order 882

directly verified on a finite $d = 5$ representation. Finiteness of the $k \in \{2, 4, 8\}$ images (this minimal-rank space) is read off from Freedman–Larsen–Wang [2] (the $k = 8$ borderline case through the $r = 10$, $n \geq 5$ clause of the same theorem), and the dense classification generally, not established here.

ACKNOWLEDGMENTS

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. The author declares no competing interests.

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